



Sports Injury Risk Detection from Video

AI-powered athlete movement analysis platform

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"Our objective is to detect these biomechanical risk factors directly from standard video footage before an injury happens."

Mitigating Costly and Preventable Sports Injuries

The Problem

Sports injuries disrupt team performance, sideline key athletes, and incur significant financial costs for organizations.

Many of these injuries stem from subtle biomechanical flaws that go unnoticed until failure occurs.

Proactive Prevention

By identifying dangerous movement patterns early, we enable proactive intervention, keeping athletes healthy and on the field.

AI-Driven Biomechanical Analysis

& Risk Prediction



Biomechanical Analytics Engine

Real-time skeletal path mapping and joint angle tracking. Identifies posture tilt, velocity vectors, and flexion angles without markers.

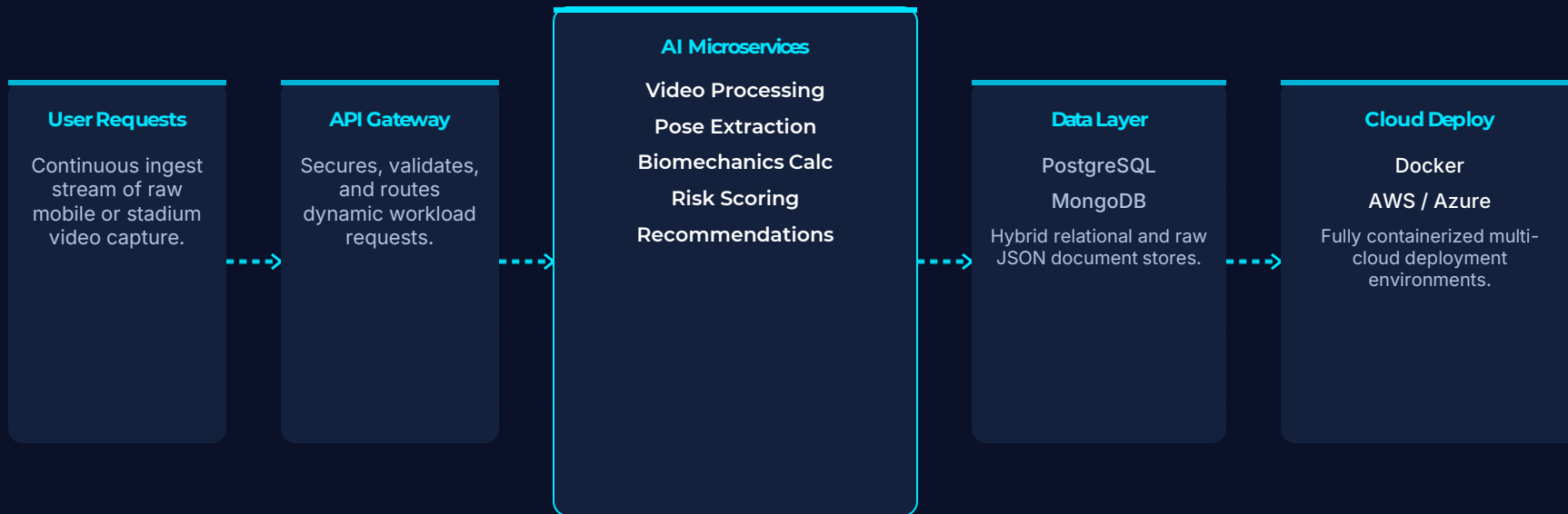
Advanced Motion Intelligence

A comprehensive platform combining Computer Vision, Pose Estimation, Biomechanics, and Machine Learning to analyze athlete movement.

- ✓ **Automated, Non-Invasive Risk Prediction**
Predicts injury risk precisely without requiring expensive or intrusive wearable sensors.
- ✓ **Data-Driven Actionable Insights**
Delivers precise directives directly to athletes, coaches, and physios for training load optimization, form correction, and recovery.



Scalable Cloud-Based AI Architecture



The platform utilizes a robust **microservices architecture** for elite scalability and real-time performance. The AI/ML processing pipeline extracts precise body metrics, stores relational and document-based data securely, and is optimized for containerized deployments across modern cloud ecosystems.

Comprehensive Modules for End-to-End Analysis



Core AI Processing Engines

Advanced AI-driven modules designed to process raw performance inputs and extract high-fidelity physical indicators:

- **Video Processing:** Efficiently ingests raw athletic footage and prepares frames for neural processing.
- **Pose Estimation:** Leverages advanced computer vision models to track spatial keypoints on the athlete's body.
- **Biomechanical Analysis:** Translates raw coordinate metrics into actionable insights regarding movement efficiency.

Foundational Layers

Ensures enterprise-grade security and structured tracking across the platform:

- Secure **Authentication & RBAC** policies safeguard sensitive organizational records.
- Detailed **Athlete Profiles** compile medical records and physical histories.

Actionable Insights & Delivery

Extracts vital downstream intelligence and distributes it to stakeholders:

- Calculates **Injury Risk Prediction** and highlights form **Anomalies**.
- Delivers real-time **Corrective Recommendations** via unified dashboards, notifications, and reports.



PROPRIETARY RISK SCORING MODEL

Injury Risk Prediction Formula

35%

BIOMECHANICS

20%

INJURY HISTORY

20%

ASYMMETRY

15%

TRAINING LOAD

Weighted Scoring Model

The score is calculated using a weighted formula that synthesizes multiple data points. Biomechanical Deviations carry the most weight (35%), followed by Historical Injury Factors (20%) and Movement Asymmetry (20%), and Training Load (15%).

Fatigue indicators complete the profile at **10%**, providing a holistic view of athlete vulnerability.

Actionable Risk Levels

LOW RISK LEVEL

MODERATE RISK LEVEL

HIGH RISK LEVEL

CRITICAL RISK LEVEL



Tailored Analytics for Every Stakeholder

The platform delivers customized views and functionality based on specific user roles to maximize utility.

✓ Athletes

Athletes can view their personal risk trends and corrective exercises.

Risk Index Trend

24%

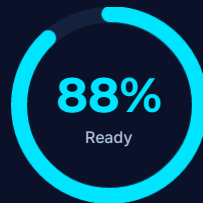
Decreasing Trend



✓ Coaches

Coaches receive team-wide overviews and readiness scores.

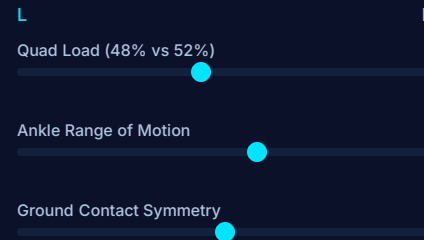
Team Readiness Score



✓ Specialists

Physios access deep biomechanics, Sports Scientists analyze long-term trends, and Admins manage secure data.

L/R Biomechanical Balance





Modern Technology Stack for High-Performance AI

The system is built on a modern, high-performance technology stack.



WEB & API

**Python / FastAPI /
React**

The backend is powered by Python and FastAPI, with a responsive React.js frontend.



DATABASES

**PostgreSQL /
MongoDB**

Data management leverages PostgreSQL for relational data and MongoDB for flexible document storage.



AI & INFRASTRUCTURE

**OpenCV / YOLOv8 /
Docker**

The core AI and Computer Vision engine utilizes MediaPipe, OpenCV, YOLOv8, and TensorFlow/PyTorch for state-of-the-art pose extraction. Infrastructure is managed via Docker and deployed on AWS or Azure.

Accelerated 8-Week Development Roadmap



We have a structured 8-week plan to deliver a production-ready platform.



Driving Proactive Prevention & Peak Performance



Shift to Proactive Prevention

The ultimate impact of this platform is a **fundamental shift** from reactive treatment to proactive injury prevention. By utilizing early warning indicators, teams can improve overall athletic performance and significantly reduce lost time due to injury.

The final deliverable is a **fully production-deployed platform** that empowers athletes and sports organizations to safeguard their most valuable assets through advanced AI and biomechanical analysis.